Declaration of Interests

Manuscript: **Fragile Authentication of PDF Signatures via Bit-Exact Quantization Index Modulation on Vector Geometry**

Journal: **Digital Signal Processing**

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

For transparency: a Ukrainian patent application (No. a 2026 06266) covering the document-sealing method described in the manuscript is pending. This is an academic priority filing by the author, not a commercial competing interest, and it does not affect the evaluation of the manuscript.

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